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B.TECH
(SEM IV) THEORY EXAMINATION 2018-19
INTRODUCTION TO MICROPROCESSOR

Time: 3 Hours ***Total Marks: 70***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

a. What are the functions of an accumulator?
b. Calculate the number of memory chips needed to design 128K-Byte memory if the Memory chip size is 2048 x 1.
c. Explain CALL & RET instructions used in 8085.
d. Explain the functions of the pins HOLD and HLDA in 8085 microprocessor.
e. What do you understand by logical address and physical address?
f. Draw the Flag register of 8086.
g. What do you mean by pipelining?

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SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

a. Explain the evolution of microprocessor with its different generations in detail.
b. Write instructions to load two unsigned numbers in register B and register C, respectively. Subtract (C) from (B). If the result is in 2's complement, convert the result in absolute magnitude and display it at PORT1; otherwise, display the positive result. (Assume (B) =42H and (C) =69H).
c. What is the need of counters and time delays? Calculate the maximum time delay that can be produced by using a register pair.
d. Write an assembly language program for the addition of two BCD numbers stored at memory location starting from XX40H, store the result at memory locations starting from XX90H.
e. Explain the various modes of operation of 8254/53 with examples.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Connect 8k byte EPROM with microprocessor 8085. The IC available is 2k × 8 EPROM, also draw its address decoding table.
(b) Draw and explain the architecture of 8085 microprocessor, also explain the programmer's model of 8085.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Draw and explain the timing diagram of opcode fetch machine cycle.
(b) Define instruction and instruction cycle. Classify the instruction set of 8085 in different groups. Explain each group with two examples; also explain the functions of the examples.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) What are interrupts? Give the classification of interrupts. Explain the hardware and software interrupts used in 8085.

(b) Calculate the 16-bit count to be loaded in register DE to obtain the loop delay of two seconds in LOOP2 (assume the clock frequency of the system to be 2MHz)

```
        MVI B, 14H
LOOP2:  LXI D, COUNT
LOOP1:  DCX D
        MOV A, D
        ORA E
        JNZ  LOOP1
        DCR B
        JNZ  LOOP2
```

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write an assembly language program to convert a binary number stored at memory location XX20H, into its equivalent ASCII-Hex code, store the codes at memory locations XX30H and XX31H.

(b) Write an assembly language program to convert $(56)_{BCD}$ to its equivalent binary number. The BCD number is stored at memory location XX50H, store the result in memory location XX60H.

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) Draw and explain the functional block diagram of 8259 PIC. Also explain its Initialization Command Words.

(b) Draw and explain the architecture of 8086 microprocessor. Calculate the physical address if CS=9105H and IP=1724H.

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